Chapter 3. Simple Linear Regression

The model, least squares, properties of the fitted line, decomposition of variation, correlation

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3.1 Regression and the straight-line model

Regression studies how a **response** Y changes with **explanatory** variables. Galton observed that sons of unusually tall or short fathers are closer to the average: heights "regress" to the mean. In Pearson's father–son data the slope is about 0.5: a father 6 inches above average has sons about 3 inches above average.

With one explanatory variable and a linear relationship we speak of **simple linear regression**.

Definition (Simple linear regression model)

For data $(x_1, y_1), \ldots, (x_n, y_n)$,

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \qquad \epsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2), \qquad i = 1, \ldots, n,$$

where the x_i are treated as fixed (nonrandom) constants and $\beta_0, \beta_1, \sigma^2$ are unknown.

The model packs three assumptions.

(1) **Linearity**: $E(Y \mid X = x) = \beta_0 + \beta_1 x$.

(2) **Normality and equal variance**: given $X = x$, $Y \sim N(\beta_0 + \beta_1 x, \sigma^2)$, with σ^2 not depending on x (**homoscedasticity**); otherwise the errors are **heteroscedastic** (Chapter 12).

(3) **Independence**: the errors are uncorrelated, $\mathrm{Cov}(\epsilon_i, \epsilon_j) = 0$ for $i \neq j$ (independent under normality).

Linear *in the parameters*: $\beta_0 + \beta_1 x + \beta_2 x^2$ is linear. For random X the analysis is conditional on the x_i: the **regression of** Y **on** X, not of X on Y.

It is often convenient to centre the explanatory variable:

$$y_i = \beta_0^* + \beta_1 (x_i - \bar{x}) + \epsilon_i, \qquad \beta_0^* = \beta_0 + \beta_1 \bar{x}. \tag{3.1}$$

3.2 Least squares

Motivation. Least squares minimizes squared vertical deviations, with closed-form estimates equal to the MLE under normal errors.

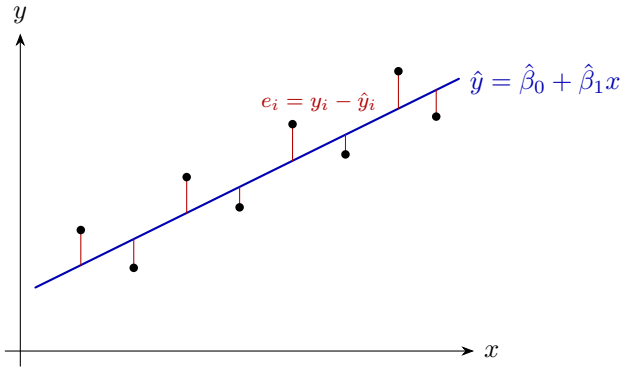


Figure. *Least squares chooses the line minimizing the sum of the squared vertical deviations e_i (red), not perpendicular distances.*

Definition (Least squares estimator)

The **least squares estimators** (LSE) $\hat\beta_0, \hat\beta_1$ minimize

$$S(\beta_0, \beta_1) = \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2.$$

Write

$$S_{xx} = \sum (x_i - \bar x)^2, \qquad S_{yy} = \sum (y_i - \bar y)^2, \qquad S_{xy} = \sum (x_i - \bar x)(y_i - \bar y).$$

Theorem 3.1 (Normal equations)

If $S_{xx} > 0$, the LSE is the unique solution of the **normal equations**

$$n\hat\beta_0 + \hat\beta_1 \sum x_i = \sum y_i, \qquad \hat\beta_0 \sum x_i + \hat\beta_1 \sum x_i^2 = \sum x_i y_i,$$

namely

$$\hat\beta_1 = \frac{S_{xy}}{S_{xx}}, \qquad \hat\beta_0 = \bar y - \hat\beta_1 \bar x.$$

Proof. Setting the partial derivatives to zero,

$$\frac{\partial S}{\partial \beta_0} = -2 \sum (y_i - \beta_0 - \beta_1 x_i) = 0, \qquad \frac{\partial S}{\partial \beta_1} = -2 \sum x_i (y_i - \beta_0 - \beta_1 x_i) = 0,$$

gives the normal equations. The first gives $\hat\beta_0 = \bar y - \hat\beta_1 \bar x$; substituting into the second, $\hat\beta_1 (\sum x_i^2 - n\bar x^2) = \sum x_i y_i - n \bar x \bar y$, i.e. $\hat\beta_1 S_{xx} = S_{xy}$. The Hessian

$$J = 2 \begin{pmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix}$$

has $n > 0$ and determinant $4nS_{xx} > 0$, so it is positive definite. S is a convex quadratic, and the stationary point is its unique minimum. ∎

In the centred form (3.1) the normal equations decouple: $\hat\beta_0^* = \bar y$ and $\hat\beta_1 = S_{xy}/S_{xx}$, so the fitted line is

$$\hat y = \bar y + \hat\beta_1 (x - \bar x).$$

Example 3.1

Advertising spending x and sales y for six stores:

x	2	3	5	6	8	10
y	5	7	8	11	13	16

Here $\bar x = 34/6 = 5.667$, $\bar y = 10$, $S_{xx} = 238 - 34^2/6 = 45.33$ and $S_{xy} = 401 - 34 \cdot 60/6 = 61$. So $\hat\beta_1 = 61/45.33 = 1.346$, $\hat\beta_0 = 10 - 1.346 \cdot 5.667 = 2.37$, and $\hat y = 2.37 + 1.346x$.

Maximum likelihood. Under normal errors the log-likelihood is

$$l(\beta_0, \beta_1, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} S(\beta_0, \beta_1).$$

For each σ^2 it is maximized by minimizing S, so the MLEs of β_0, β_1 are the LSEs, and $\hat\sigma^2_{\text{MLE}} = S(\hat\beta_0, \hat\beta_1)/n$.

3.3 Properties of the fitted line

The **fitted values** are $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ and the **residuals** are $e_i = y_i - \hat{y}_i$.

Theorem 3.2

(a) $\sum e_i = 0$; equivalently $\sum y_i = \sum \hat{y}_i$.

(b) $\sum x_i e_i = 0$.

(c) $\sum \hat{y}_i e_i = 0$.

(d) The fitted line passes through $(\bar{x}, \bar{y})$.

(e) $\sum e_i^2$ is the minimum of $S(\beta_0, \beta_1)$.

Proof. (a) and (b) are the two normal equations, since $\sum e_i = \sum(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)$ and $\sum x_i e_i = \sum x_i(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)$.

(c) $\sum \hat{y}_i e_i = \hat{\beta}_0 \sum e_i + \hat{\beta}_1 \sum x_i e_i = 0$.

(d) $\hat{\beta}_0 + \hat{\beta}_1 \bar{x} = \bar{y}$.

(e) is the definition of the LSE. ■

Geometrically the residual vector is orthogonal to $\mathbf{1}$, $\mathbf{x}$ and the fitted vector, so e_i against $\hat{y}_i$ (or x_i) shows no linear trend.

3.4 Decomposition of variation

Motivation. SST splits into regression and residual parts; R^2 is their ratio.

Theorem 3.3 (Sum of squares decomposition)

$$\underbrace{\sum (y_i - \bar{y})^2}_{SST} = \underbrace{\sum (y_i - \hat{y}_i)^2}_{SSE} + \underbrace{\sum (\hat{y}_i - \bar{y})^2}_{SSR},$$

and $SSR = \hat{\beta}_1^2 S_{xx} = S_{xy}^2 / S_{xx}$.

Proof. Write $y_i - \bar{y} = (y_i - \hat{y}_i) + (\hat{y}_i - \bar{y})$ and square. The cross term is $2\sum e_i(\hat{y}_i - \bar{y}) = 2\sum \hat{y}_i e_i - 2\bar{y}\sum e_i = 0$ by Theorem 3.2. Since $\hat{y}_i - \bar{y} = \hat{\beta}_1(x_i - \bar{x})$, $SSR = \hat{\beta}_1^2 S_{xx} = S_{xy}^2 / S_{xx}$. ■

SST, SSE, SSR are the **total**, **residual** (error) and **regression** sums of squares, with $n-1$, $n-2$, 1 degrees of freedom.

Definition (Coefficient of determination)

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST} \in [0, 1].$$

R^2 is the proportion of variation explained: 1 for a perfect fit, 0 for $\hat{\beta}_1 = 0$. The standard error of estimate is $s = \sqrt{MSE}$, $MSE = SSE/(n-2)$, unbiased for σ^2 (Chapter 4).

Example 3.2

In Example 3.1, $\sum y_i^2 = 684$, so $SST = S_{yy} = 684 - 600 = 84$. Then $SSR = 61^2/45.33 = 82.08$, $SSE = 84 - 82.08 = 1.92$ and $R^2 = 0.977$: advertising explains about 98% of the variation in sales.

3.5 Correlation

For a random pair (X, Y) the **population correlation** is $\rho_{XY} = \mathrm{Cov}(X, Y)/\sqrt{\mathrm{Var}\, X\, \mathrm{Var}\, Y}$, and the **sample correlation** is

$$r_{xy} = \frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}} \in [-1, 1].$$

Theorem 3.4

(a) $R^2 = r_{xy}^2$ in simple linear regression.

(b) $r_{xy} = \hat{\beta}_1\, s_x/s_y$, where $s_x^2 = S_{xx}/(n-1)$ and $s_y^2 = S_{yy}/(n-1)$; hence r_{xy} has the sign of $\hat{\beta}_1$.

Proof. (a) By Theorem 3.3, $R^2 = \frac{SSR}{SST} = \frac{S_{xy}^2}{S_{xx} S_{yy}} = r_{xy}^2$.

(b) $r_{xy} = \frac{S_{xy}}{S_{xx}} \sqrt{\frac{S_{xx}}{S_{yy}}} = \hat{\beta}_1 \frac{s_x}{s_y}$. ■

r_{xy} measures only *linear* association. For $x = 0, 1, 2, 3, 4$ and $y = x^2 - 4x + 6 = (6, 3, 2, 3, 6)$, $S_{xy} = 0$ and $r_{xy} = 0$, although y is a deterministic function of x.

Remark

$\hat{\beta}_1$ is the expected change in y per unit x; r_{xy} is unit-free and symmetric. The regression of x on y has slope S_{xy}/S_{yy}; the slopes multiply to $r_{xy}^2 \le 1$, so the lines coincide only if $|r_{xy}| = 1$.

3.6 Selected exercises

Motivation. Least squares on special designs: through the origin, two distinct x values, and the slope as an average of pairwise slopes.

Exercise 3.1 (Regression through the origin)

Consider $y_i = \beta x_i + \epsilon_i$, $i = 1, \ldots, n$, with $\sum x_i^2 > 0$ and no intercept.
(a) Show that the least squares estimator is $\hat{\beta} = \sum x_i y_i / \sum x_i^2$ and that the residuals $e_i = y_i - \hat{\beta} x_i$ satisfy $\sum x_i e_i = 0$, but in general $\sum e_i \neq 0$.
(b) Show that $\sum y_i^2 = \sum \hat{y}_i^2 + \sum e_i^2$, and give an example in which the centred decomposition $SST = SSR + SSE$ of Theorem 3.3 fails.

Solution [Supplement].

(a) $S(\beta) = \sum (y_i - \beta x_i)^2$ has $S'(\beta) = -2 \sum x_i (y_i - \beta x_i)$, $S'' = 2 \sum x_i^2 > 0$: strictly convex, minimized where $\sum x_i e_i = 0$. Nothing forces $\sum e_i = 0$: $x = (1, 2)$, $y = (1, 1)$ give $\hat{\beta} = 3/5$, $e = (0.4, -0.2)$, $\sum e_i = 0.2$.

(b) Since $\hat{y}_i = \hat{\beta} x_i$,

$$\sum y_i^2 = \sum (\hat{y}_i + e_i)^2 = \sum \hat{y}_i^2 + 2\hat{\beta} \sum x_i e_i + \sum e_i^2 = \sum \hat{y}_i^2 + \sum e_i^2.$$

There $\bar{y} = 1$, $SST = 0$, $SSE = 0.2 > 0$: $SST \neq SSR + SSE$ and R^2 is meaningless. Without intercept use the uncorrected $\sum y_i^2$, $\sum \hat{y}_i^2$. ■

Exercise 3.2 (Two design points)

Let the x_i take two values $a < b$, $n_a, n_b \geq 1$ times ($n = n_a + n_b$), with response means $\bar{y}_a$, $\bar{y}_b$. Show that

$$S_{xx} = \frac{n_a n_b (b-a)^2}{n}, \qquad \hat{\beta}_1 = \frac{\bar{y}_b - \bar{y}_a}{b-a},$$

and that the fitted line passes through $(a, \bar{y}_a)$ and $(b, \bar{y}_b)$.

Solution [Supplement].

Step 1. $\bar{x} = (n_a a + n_b b)/n$, so $a - \bar{x} = -n_b(b-a)/n$ and $b - \bar{x} = n_a(b-a)/n$.

Step 2. Hence

$$S_{xx} = n_a \frac{n_b^2 (b-a)^2}{n^2} + n_b \frac{n_a^2 (b-a)^2}{n^2} = \frac{n_a n_b (b-a)^2 (n_a + n_b)}{n^2} = \frac{n_a n_b (b-a)^2}{n}.$$

Step 3. Using $S_{xy} = \sum (x_i - \bar{x}) y_i$ and grouping the observations by their x value,

$$S_{xy} = -\frac{n_b(b-a)}{n} n_a \bar{y}_a + \frac{n_a(b-a)}{n} n_b \bar{y}_b = \frac{n_a n_b (b-a)(\bar{y}_b - \bar{y}_a)}{n},$$

and $\hat{\beta}_1 = S_{xy}/S_{xx} = (\bar{y}_b - \bar{y}_a)/(b-a)$.

Step 4. The fitted line is $\hat{y}(x) = \bar{y} + \hat{\beta}_1 (x - \bar{x})$ with $\bar{y} = (n_a \bar{y}_a + n_b \bar{y}_b)/n$. At $x = a$, by Step 1,

$$\hat{y}(a) = \frac{n_a \bar{y}_a + n_b \bar{y}_b}{n} - \frac{n_b (\bar{y}_b - \bar{y}_a)}{n} = \frac{(n_a + n_b)\bar{y}_a}{n} = \bar{y}_a,$$

and in the same way $\hat{y}(b) = \bar{y}_b$. So the line joins the two group means, and all the variation within the groups goes into SSE. ■

Exercise 3.3 (The slope as an average of pairwise slopes)

Assume $S_{xx} > 0$. For $i < j$ with $x_i \neq x_j$, let $s_{ij} = (y_j - y_i)/(x_j - x_i)$ be the slope of the line through observations i and j. Show that

$$\hat{\beta}_1 = \sum_{i<j,\ x_i \neq x_j} w_{ij} s_{ij}, \qquad w_{ij} = \frac{(x_j - x_i)^2}{\sum_{k<l} (x_l - x_k)^2},$$

a weighted average of the pairwise slopes with weights proportional to the squared distance between the x values.

Solution [Supplement].

Step 1. For any numbers u_i, v_i, expanding the double sum,

$$\sum_{i<j} (u_j - u_i)(v_j - v_i) = \frac{1}{2} \sum_{i,j} (u_j - u_i)(v_j - v_i) = n \sum_i u_i v_i - \Big(\sum_i u_i \Big) \Big(\sum_i v_i \Big) = n \sum_i (u_i - \bar{u})(v_i - \bar{v}).$$

Step 2. With $u = v = x$ this gives $\sum_{k<l}(x_l - x_k)^2 = nS_{xx} > 0$, so the weights are well defined, nonnegative and sum to 1 (pairs with $x_i = x_j$ have zero weight). With $u = x$, $v = y$ it gives $\sum_{i<j}(x_j - x_i)(y_j - y_i) = nS_{xy}$.

Step 3. For a pair with $x_i \neq x_j$, $(x_j - x_i)(y_j - y_i) = (x_j - x_i)^2 s_{ij}$, and pairs with $x_i = x_j$ contribute 0 to the sum in Step 2. Therefore

$$\sum_{i<j,\ x_i \neq x_j} w_{ij} s_{ij} = \frac{\sum_{i<j}(x_j - x_i)(y_j - y_i)}{nS_{xx}} = \frac{nS_{xy}}{nS_{xx}} = \hat{\beta}_1.$$

Pairs of points far apart in x determine their slope more reliably, and least squares gives them more weight. ■